

SQL Assignment - Basic

We are analyzing a small online shopping platform where customers register and place orders. The company wants to track customer details and their purchases for reporting and analysis.

- Database Name: **RetailDB_1**
- Purpose: To understand who the customers are, when they joined, and what orders they placed.
- At this stage, we are only storing customer information and their orders.

Tables in RetailDB:

1. **Customers** – stores details of each registered customer.

(**customer_id**, name, email, city, signup_date)

2. **Products** - holds all the products details

(**product_id**, product_name, category, price)

3. **Orders** – keeps records of every order placed, linked to the customer.

(**order_id**, customer_id, product_id, order_date, quantity, total_amount, payment_mode)

After inserting some data, the Tables looks like:

Customers

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Delhi	2024-01-15
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2024-02-20
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2024-03-05
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2024-03-22
5	Karan Singh	karan.singh@outlook.com	Chennai	2024-04-10
6	Neha Verma	neha.verma@gmail.com	Pune	2024-05-08
7	Arjun Mehta	arjun.mehta@gmail.com	Jaipur	2024-05-15
8	Ananya Iyer	ananya.iyer@gmail.com	Bengaluru	2024-06-02
9	Vikram Das	vikram.das@gmail.com	Kolkata	2024-06-20
10	Meera Nair	meera.nair@gmail.com	Kochi	2024-07-01

Products

product_id	product_name	category	price
1	iPhone 15	Electronics	79999.00
2	Samsung Galaxy S24	Electronics	74999.00
3	Noise Smartwatch	Wearables	4999.00
4	Boat Earbuds	Wearables	2499.00
5	Kurta Set	Fashion	1999.00
6	Running Shoes	Fashion	2999.00
7	Prestige Mixer Grinder	Home Appliances	6499.00
8	Tata Tea 1kg Pack	Groceries	499.00
9	Amul Butter 500g	Groceries	285.00
10	Sony Bravia 55" TV	Electronics	59999.00

Orders

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
1	1	1	2024-07-10	1	79999.00	UPI
2	2	2	2024-07-15	1	74999.00	Credit Card
3	3	3	2024-07-20	1	4999.00	Net Banking
4	4	6	2024-07-22	2	5998.00	Debit Card
5	5	5	2024-08-01	1	1999.00	UPI
6	6	7	2024-08-03	1	6499.00	Credit Card
7	8	10	2024-08-05	1	59999.00	Debit Card
8	9	8	2024-08-10	3	1497.00	UPI

Instructions to Load the tables on MySQL Workbench:

1. Download the files from folder link:

https://drive.google.com/drive/folders/1gK2lirWNX087OWMAbn2am1A7CTcJvvPR?usp=drive_link

2. Create the database with name as - **RetailDB_1**, with the following table as:
(just copy paste on sql script)

```
CREATE TABLE Customers (  
    customer_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(150) UNIQUE,  
    city VARCHAR(50),  
    signup_date DATE  
);
```

```
CREATE TABLE Products (  
    product_id INT AUTO_INCREMENT PRIMARY KEY,  
    product_name VARCHAR(100),
```

```
category VARCHAR(50),  
price DECIMAL(10,2)  
);  
  
CREATE TABLE Orders (  
    order_id INT AUTO_INCREMENT PRIMARY KEY,  
    customer_id INT,  
    product_id INT,  
    order_date DATE,  
    quantity INT,  
    total_amount DECIMAL(10,2),  
    payment_mode VARCHAR(50),  
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),  
    FOREIGN KEY (product_id) REFERENCES Products(product_id)  
);
```

3. Load the files on the created tables in MySQL Workbench using fetch csv data.
4. Can view with select command.

Task: Solve the below mentioned questions by writing SQL queries.

1. Fetch all customers from the database.
2. Show only the customer names and their cities.
3. Find customers who live in Mumbai.
4. Get all orders placed after 1st August 2024.
5. List all products priced greater than ₹5000.

6. Count how many customers exist in the system.
7. Update a customer's city (e.g., change Rohit Kumar's city to Hyderabad).
8. Delete an order (e.g., remove order with ID = 5).
9. Display product names with their original price and price increased by 10%.
10. Show only the unique cities where customers live.
11. Get the first 3 customers who signed up.
12. Skip the first 2 customers and fetch the next 3 customers.
13. Find products with prices between ₹2000 and ₹6000.
14. Find customers who are from Mumbai OR Chennai.
15. Find customers who are NOT from Delhi.
16. Find orders that are NOT paid by UPI.
17. Get the average order amount across all orders.
18. Show the highest order amount.
19. Show the lowest product price.
20. Find the total money spent across all orders.